

# Choupo

The Open Source Chemical Process Simulator

# Explorer Guide

Choupo-dev

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# 1 What the Explorer is

The Explorer is Choupo's *interactive thermodynamics bench*: pick components, pick a view, and SEE what the selected property models do — instantly, in the browser, with no case authoring at all. It answers the question that comes *before* any simulation: **"what does this thermodynamics look like?"**

It sits deliberately between two other surfaces:

- The **props bench** (`choupoProps` cases) asks the same questions *reproducibly*: a case file, golden-locked numbers, validation blocks. The Explorer is its sketchpad — what you find here, you pin down there.
- The **simulators** consume the models the Explorer displays. A surprise in a flowsheet (an azeotrope, a scaling mineral, a runaway $\gamma$) is usually visible in the Explorer in thirty seconds.

Everything the Explorer draws is computed by the SAME C++ engine (compiled to WebAssembly) with the SAME curated catalogue (`data/standards/`) the solvers use. There is no "display model": if the Explorer shows an azeotrope at $x = 0.89$, the distillation column will hit it at $x = 0.89$.

# 2 The workflow

1. **Pick components** in the left panel. The list is the live catalogue; each card shows the data the component actually carries (criticals, $P^{\text{sat}}$, $c_p$, formation — and what it lacks). Case-local components (from an open case's `constant/`) appear alongside, marked.
2. **Pick a view** (the tabs enable themselves when their component-count and data requirements are met; a disabled tab tells you WHY it is disabled).
3. **Pick the method** where more than one applies — activity model, equation of state, $P^{\text{sat}}$ correlation. Overlaying two methods on the same axes is the fastest modelling lesson Choupo offers: the gap between the curves IS the modelling uncertainty.
4. **Export**: every view is a Plotly figure (zoom, hover, save PNG) backed by numbers you can carry into a props case for the golden-locked version.

# 3 The views, one by one

## 3.1 The compound browser (the Explorer's landing)

The catalogue is a *tree*, not a list. Groups derived from each record's own facts fold and unfold with their member count; the groups a student charges to a vessel open by default, the combustion library, the radicals and the synthetic test stand-ins start folded. A group too large for one screen is subdivided into *families*, and the family comes from the record, never from the name: first the *declared UNIFAC groups* (the highest-precedence group present decides — an acid before an ester before an alcohol, down to an alkane); only for a record with no group decomposition, the *elemental formula*, coarser and labelled so with a "(by formula)" suffix. A record with neither is shown under *"No formula declared in the record"* — a visible curation gap, never a guess. Typing in the search box flattens the tree; a double click inspects a component without changing the set. The browser reads the catalogue; it does not edit a record.

The browser is the left half of the Explorer's *landing* — the screen `?workspace=explore` opens; its right half shows the record of whichever compound you last touched — identity, what the record carries, the curation dossier's verdict — and its **Explore properties** button opens the property surfaces in a tab of their own, carrying the set in the address (`?workspace=properties&components=water,ethanol`), which makes an exploration a link you can hand to somebody. Neither tab needs the other: the surfaces have their own door on the hub, keep a compound rail, and open perfectly well with nothing selected, and a component the address names but the catalogue cannot resolve is left out of the set and said so, never dropped in silence.

## 3.2 Property vs $T/P$ (scan)

Any pure-component property — $P^{\mathrm{sat}}$, liquid density, $c_p$, viscosity, enthalpy — swept over temperature or pressure, for one or many components. Use it to: check a correlation's range before trusting it in a case; compare components; spot the nonsense that extrapolation produces outside a fit's window (the guide's recurring lesson: a correlation is DATA, and data has a domain).

## 3.3 Pure phase diagram ($P$–$T$)

The full phase map of one substance: sublimation, melting and vapour-pressure lines meeting at the triple point, ending at the critical point. Expect water's negatively-sloped melting line (the anomaly), and the supercritical region where the liquid/vapour distinction dissolves.

## 3.4 Binary boiling envelope ($T$–$x$–$y$)

The bubble and dew curves of a binary at fixed $P$. This is where azeotropes announce themselves: the curves touch, and no distillation can cross the touching point. Switch the activity model (ideal → NRTL → UNIFAC) and watch the envelope bend — ideal solutions cannot azeotrope; real ones do.

## 3.5 $\gamma(x)$ — activity coefficients

The raw non-ideality of a binary: $\gamma_1, \gamma_2$ against composition, with the infinite-dilution values $\gamma^{\infty}$ at the edges. $\gamma > 1$: the pair dislikes mixing (positive deviation, minimum-boiling azeotropes); $\gamma < 1$: association (maximum-boiling). The Gibbs–Duhem constraint ties the two curves together — one cannot bend without the other answering.

## 3.6 Binary flash ($x$–$y$ + lever rule)

One flash, drawn: feed, vapour and liquid on the equilibrium diagram with the lever rule made visible. Use it to understand what a single equilibrium stage can and cannot do.

## 3.7 Binary and ternary LLE

Liquid–liquid splitting from the mixing Gibbs energy: $g_{\mathrm{mix}}(x)$ with the common-tangent construction (binary), and the ternary solubility envelope with tie-lines (UNIFAC-predicted). Expect the tangent points — not the minima — to define the coexisting phases; that distinction is the whole thermodynamics of demixing.

## 3.8 Ternary boiling surface

$T_{\mathrm{bubble}}$ over the composition triangle: valleys are distillation boundaries, and azeotropes sit at the stationary points.

## 3.9 Scaling (SI vs recovery)

The membrane/brine screening view: concentrate a water composition over recovery and watch each mineral's saturation index climb toward SI = 0 (the precipitation line). The Davies-vs-Pitzer method switch is the lesson: at high ionic strength the two activity models give DIFFERENT scaling verdicts, and choosing one is an engineering decision.

## 3.10 Equilibrium map (Gibbs)

The chemical-reaction view: pick two or more gas-phase species (with plain formulas — the atom matrix is derived from them, glass-box) and a feed, and get iso-lines of a product's equilibrium mole fraction over the $T \times P$ plane. The whole point is the tension it makes visible: for ammonia ($N_2 + 3H_2 \rightleftharpoons 2NH_3$) the equilibrium optimum sits cold and high-pressure, yet the labelled industrial window sits hot — where the catalyst can work. Answers the beginner's real confusion first ("where do I write the reaction?" — you don't; Gibbs redistributes the ATOMS you fed, shown as a live element balance). *Click any cell* for its full equilibrium composition and the ready-to-run `gibbsReactor` case dict — the map is a case generator. The *advanced* toggle exposes the temperature-approach $\Delta T$ (the empirical closeness-to-equilibrium knob commercial tools hide); it snaps rather than animates, and ghosts the $\Delta T = 0$ contours underneath so you read the before/after, not a motion. Unconverged cells are marked, never interpolated.

## 3.11 Steam tables (IF97)

Saturation and superheat properties of water from IAPWS-IF97 — $h_f, h_g, h_{fg}$, entropies, volumes — the backbone of every steam and power-cycle case, browsable.

## 3.12 EduTools (the Methods workspace)

Since 2026-08-15 the classical *method constructions* live in their own workspace, **EduTools**, split from the Explorer by one criterion: *a property surface answers "what does this system's property look like?"* (the Explorer); *a method construction answers "what does the classical graphical method say about a design question?"* — McCabe–Thiele, Kremser, Rayleigh, $\varepsilon$-NTU, pinch composites, and derivation pages for the property models. Open it from the hub's **EduTools** button: it is a tab of its own, needs no case, and does not talk to the flowsheet. A tool is chosen from the top row of the tab; every tool carries a one-line "teaches" caption and its Theory Guide link. Every curve is an engine run (`choupoProps` in WebAssembly) over a transient synthesized case — *zero physics in TypeScript*; the staircase drawn on it is geometry. Planned tools stay visible and disabled, naming what will feed them. It never writes a case: "Author → copy propsDict" emits the dict for you to author on disk.

## 3.13 Literature ("who measured this?")

Provenance is the differentiator, and this workspace puts its question one click from the flowsheet. With a case open, the tab's menu carries **Literature**: type compound names (every word must match a compound in the article) or click a chip for one of the case's own components, and read who measured that system — authors, year, title, journal, a DOI link, and the properties measured. It searches your *private* mirror of the NIST/TRC ThermoML Archive, installed once with `bin/choupo-thermoml sync` on your own machine; the runtime never reads it and the public site has no mirror, so there it shows the install instructions instead. That absence is a status, not an empty result: *"nobody measured this"* and *"you have no mirror"* are different sentences. It is a *reading list, never a value*: no number is copied into a case or a record, and the only outbound action is opening the article. Read it and cite it — never this index.

# 4 Provenance and honesty rules

The Explorer inherits the project's credo:

- **Every number has a source.** Component cards show where each value came from; estimates carry their UNVALIDATED badge; nothing is silently filled in.
- **Disabled means explained.** A view that cannot run tells you which requirement failed (component count, missing VLE data, missing UNIFAC groups) — never a dead button.
- **The Explorer never writes.** It reads the catalogue and the open case; promoting data, fitting parameters and locking goldens belong to the props bench and the curation tools. Sketch here, commit there.

# 5 From Explorer to case: the hand-off

A typical session: you notice in the $T$–$x$–$y$ view that your solvent pair azeotropes at 78 % — so the simple column in your flowsheet cannot deliver 99 % purity. The hand-off is: (1) reproduce the envelope as a props case (`propertyScanBinary`) so the finding is golden-locked; (2) switch the flowsheet to the azeotrope-aware design (pressure-swing or entrainer); (3) cite the Explorer finding in the case header. The Explorer is where you LEARN; cases are where knowledge is kept.